MINI PROJECT-2 (22ET6PWMP2)

A Report
ON
"Surveillance Robot"

Submitted in partial fulfilment of the requirements for the award of degree of

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IN
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VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

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BONAFIDE CERTIFICATE

This is to certify that the project titled "***Surveillance Robot***" is a bonafide work carried out by **A U Shashank Kashyap(1BM21ET001) and Subodh S Katti(1BM21ET048)** in partial fulfilment for the completion of **MINI PROJECT-2 (22ET6PWMP2)** ***during*** the academic year 2023-2024.

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External Examination

Name of the Examiners Signature with date

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TITLE:

Surveillance Robot

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DATE: June 2024

Objective:

Design a surveillance robot with these key components: a robust chassis, high-resolution cameras, and a powerful processor like NVIDIA GPU RTX 3200. Use OpenCV for image processing, and machine learning models (YOLO) for object detection. Develop a remote monitoring website, and ensure secure data transmission. Test thoroughly in controlled environments before field deployment.

Description:

This report details the development and implementation of an advanced surveillance robot equipped with an ESP32-CAM module for real-time monitoring and object detection. The primary objective is to enhance surveillance capabilities through live video feeds and remote control functionalities, allowing users to navigate the robot to various locations for thorough coverage. By incorporating sophisticated computer vision algorithms, the robot can identify and track objects within its field of view, providing valuable insights and alerts. The system is designed with a user-friendly web interface hosted on the ESP32-CAM board, ensuring accessible and intuitive control from any connected device with a web browser. This comprehensive approach aims to improve security and monitoring efficiency in various applications, ranging from residential to industrial environments.

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ABSTRACT

The ESP32-CAM surveillance robot is a versatile system leveraging the ESP32-CAM board and a mobile robot chassis. This setup combines a low-cost development board with integrated camera and Wi-Fi capabilities, enabling mobile surveillance. Controlled via a web interface hosted on the ESP32CAM, users can manage the robot's movements, access live video feeds, and capture snapshots. Furthermore, the system can be programmed to employ computer vision algorithms for motion detection, object tracking, and alerting. This innovative solution finds applications in home security, remote area monitoring, and industrial surveillance, offering an affordable and user-friendly means to remotely monitor surroundings.

CHAPTER 1. INTRODUCTION

1.1. Overview

The surveillance robot using ESP32-CAM is designed to provide an affordable and efficient solution for monitoring various environments. The project leverages the ESP32-CAM module, which integrates a camera and Wi-Fi connectivity, mounted on a mobile robot chassis. This setup allows the robot to capture and transmit video footage, enabling real-time remote surveillance. Controlled through a web interface, users can manoeuvre the robot, view live video streams. The system is ideal for home security, industrial monitoring, and remote location surveillance due to its low cost, ease of use, and reliable performance in various conditions .

1.2. Problem Statement

The problem addressed by the project is the need for a cost-effective, efficient surveillance solution that can be remotely controlled and monitored. Traditional surveillance systems are often expensive, complex to set up, and lack mobility, limiting their effectiveness in dynamic environments. The project aims to solve these issues by developing a low-cost surveillance robot using the ESP32-CAM module, which integrates a camera and Wi-Fi connectivity into a small, mobile platform. This allows for flexible, real-time monitoring of various locations, providing a versatile solution for home security, industrial monitoring, and remote area surveillance

1.3. Proposed Solution

The project aims to address the need for a cost-effective, efficient, and remotely controlled surveillance solution. It proposes a versatile surveillance robot using the ESP32-CAM module which will offer real-time video feeds, remote navigation. A user-friendly web interface hosted on the ESP32-CAM board will enable easy control from any web browser. Built on a small, mobile platform with integrated Wi-Fi, this low-cost solution provides flexible deployment for home security, industrial monitoring, and remote area surveillance, overcoming the limitations of traditional surveillance systems.

CHAPTER 2. LITERATURE SURVEY

In [1], the author discusses a system for observing the human movement in the war field and border regions which can reduce the risk of human life as soldiers of armed forces can assess the condition of the territory before entering it. The war field robot comprises Arduino uno board which is fitted with L293D motor driver, HC-05 Bluetooth module and night vision wireless camera. Camera is mounted on the top of the robot and can monitor minimum of 100 m transmission distance. It has the feature of rotating 360 degress by the means of android application. That android application is created through MIT app inventor and is used for the total navigation control of the robot.

In [2], the author proposed a robot to execute the act of continuous surveillance in the domestic areas. It is capable to reduce the human labour as well as human error. The robot is controlled by mobile or laptop through Internet of Things by the means of Cayenne software which is used to send commands to the robotic system. Arduino interfaced with ESP-8266 Wi-Fi module for unlimited range of operation receives those commands. DC motors, ultrasonic sensor to detect any obstruction, IR sensors and wireless camera that provides audio and video streaming data to the user end are used in this project. Arduino Integrated Development Environment which is a cross platform application is used for writing codes for Arduino board.

In [3], the author presents a cost effective robot using Arduino Uno microcontroller which can carry out the act of surveillance as well as rescue operation. The robot is equipped with passive infrared sensor, ultrasonic ranging module, DHT11 temperature and humidity sensor, MQ-135 air quality sensor, accelerometer sensor, gyroscope sensor, magnetometer sensor, GPS sensor, Bluetooth and WIFI module, motor driver and robotic arm. The robotic arm is a type of jointed arm and consists of gripper to pick up objects with ease. It is controlled remotely by end user via internet. Solar panel is also connected to provide with the facility to charge battery with solar power.

In [4], the author proposed a surveillance robot for real time video streaming and audio transfer. The wireless mini robot is controlled by android based application named as BLYNK. Arduino UNO R3 is used as robot control board and it is incorporated with NodeMcu ESP module and L293D Servo Stepper DC Motor Shield. NodeMcu ESP8266 Wi-Fi module is paired with android mobile using username and password. Username and password is only known and predefined by the programmer in the programming of the system. It makes this wireless mini robot control completely secured from trespasser.

In [5], the author put forward the monitoring and controlling of mobile robot via internet through Raspberry Pi to control the terrorist attack throughout the world. Raspberry Pi Model B+ is used to incorporate all the sensors and motor drivers. The programming language used in the system is based on LINUX platform. Low cost PIR sensor and smoke sensor is connected to Raspberry Pi microcontroller to trace out suspicious activities, intruders and to detect fire accidents. These sensors enhance the effectiveness of the project in the case of camera failure. This camera can be rotated to get the better visibility of the surrounding. For wireless connectivity Wi-Fi dongle is used. Web page is created using HTML for navigation control.

In [6], the author discusses a spy robot which is based on Raspbian operating system. Raspberry Pi 3 Model-B is used which is 3rd generation Raspberry minicomputer with a 64-Bit 1.2 GHz quad core processor, 1GB RAM, WIFI and Bluetooth 4.1 controller. Python programming language is used to write the script for client and server communication. The Raspberry Pi is connected to the H-Bridge IC L293D to control the movement of the direction of motors based to the signals received by the end user.

CHAPTER 3. METHODOLOGY

3.1 BLOCK DIAGRAM OF PROPOSED MODEL

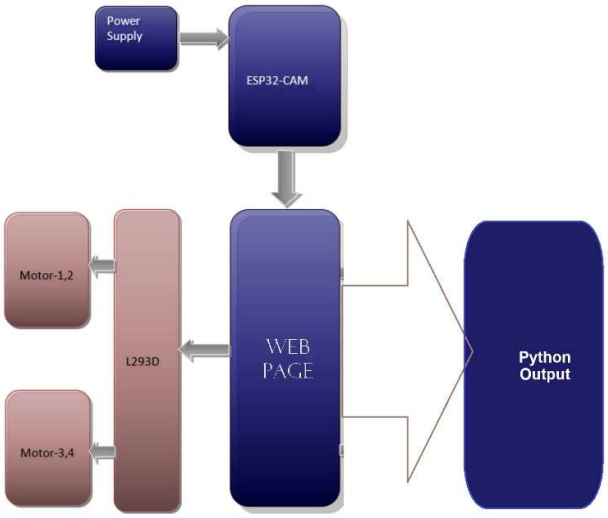


fig 3.1: Pictorial Representation of Proposed Model

The above shown figure is the block diagram of the proposed model. It consists of ESP –32 cam, L293D Motor Driver, 4-DC Motors, Power bank, Servo motors, Other Misc Components.

The ESP32-CAM surveillance robot operates by capturing video feed through its camera and streaming it over Wi-Fi to a web interface. Users can remotely control the robot's movements using this interface, sending commands to the ESP32-CAM, which then relays these commands to the L293D motor driver. The motor driver, in turn, controls the DC motors to move the robot in the desired direction. Servo motors adjust the camera angle to maintain optimal surveillance coverage.

3.2 CIRCUIT SCHEMATIC

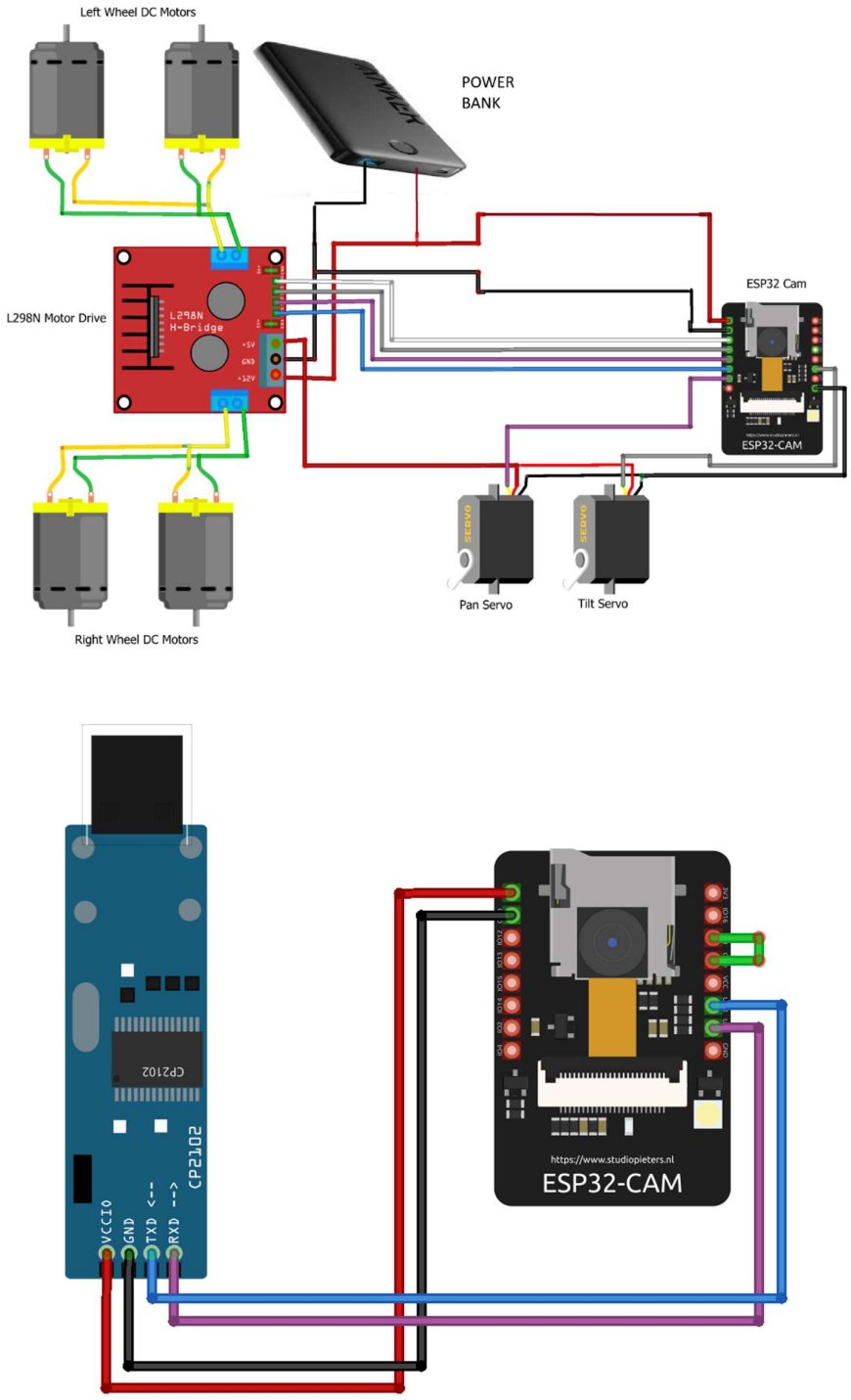


Fig3.2: Circuit Schematic

3.3 METHOD OF OPERATION

Hardware Implementation

The initial step involves assembling the necessary hardware components for the project. These components include the ESP32-CAM module, motor drivers, motors, wheels, chassis, battery, and infrared sensor. Jumper wires are used to connect the ESP32-CAM module to the motor drivers and the infrared sensor. The motors are connected to the motor drivers, with the wheels attached to the motors. The battery supplies power to both the ESP32-CAM module and the motor drivers, ensuring the robot operates effectively.

ESP32-CAM

The ESP32-CAM, is a compact camera module based on the ESP32 microcontroller and the OV2640 sensor. It can capture images, stream video, and perform various image processing tasks. Featuring Wi-Fi and Bluetooth connectivity, the ESP32-CAM is ideal for IoT and surveillance applications. Key features include the ESP32 microcontroller, a 2 MP OV2640 camera sensor, Wi-Fi and Bluetooth connectivity, GPIO pins for interfacing with other devices, a micro SD card slot for storage, and a 5 V DC power supply. Using the ESP32-CAM requires programming knowledge, particularly with the Arduino IDE, and basic electronics skills for connecting wires and components to the GPIO pins. This versatile module is suitable for projects like smart security cameras, remote-controlled cars with cameras, and face recognition systems, making it a powerful tool for makers and hobbyists.

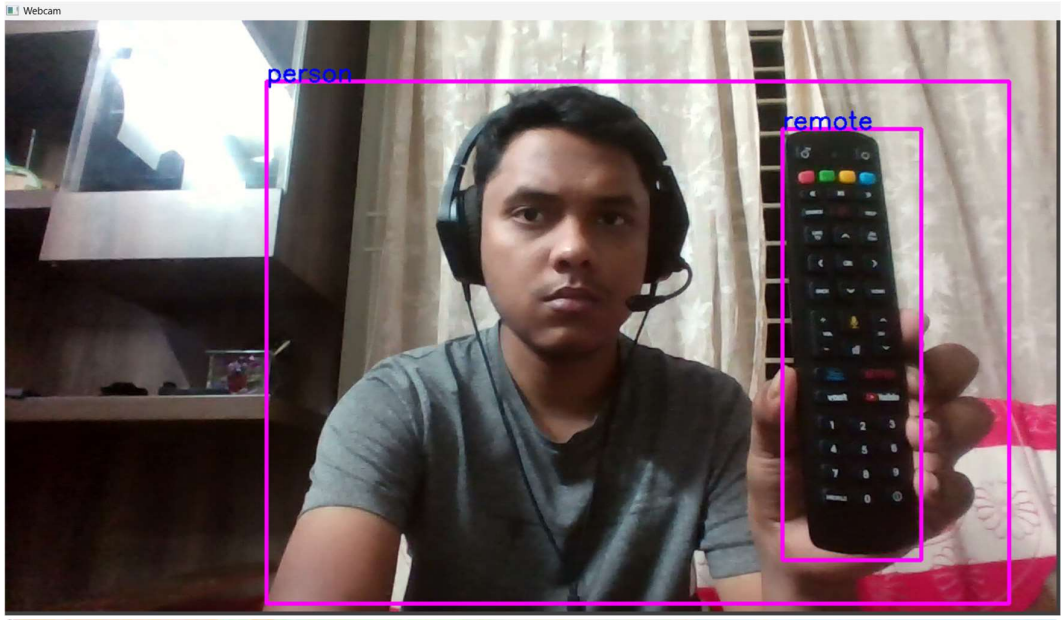
MOTOR DRIVER

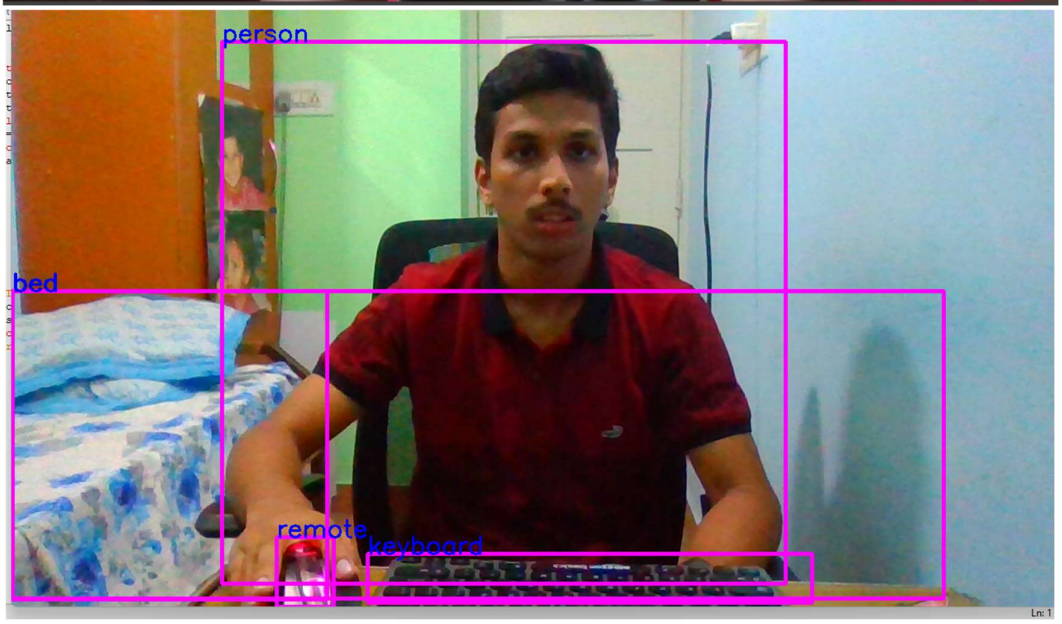
The L298N is a popular motor driver IC that controls the direction and speed of DC and stepper motors, supporting up to 2 amps of continuous current per channel and an input voltage range of 5V to 35V. It includes two H-bridge circuits, each with four transistors that can be independently controlled, allowing motors to run in either direction. To use the L298N, we need to connect it to the microcontroller or control circuitry using its various pins . ENA and ENB are enable pins for channels A and B and can be linked to digital output pins on the microcontroller. IN1, IN2, IN3, and IN4 are control pins for the H-bridge circuits, allowing for direction and speed regulation. OUT1, OUT2, OUT3, and OUT4 are the output pins that connect to the motor. We need to ensure the motor is connected to the OUT pins and the motor power supply is connected to the L298N's V+ and GND pins.

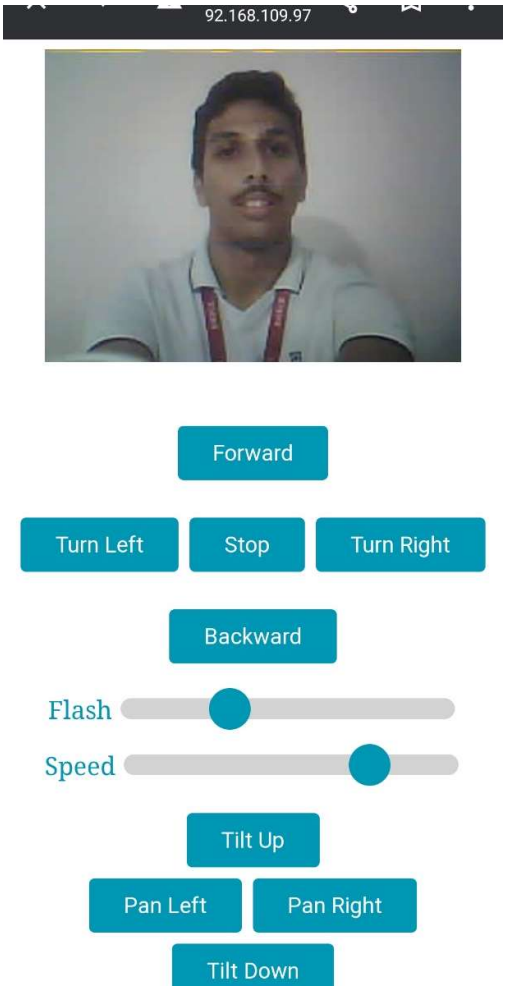
FTDI

FTDI (Future Technology Devices International) programmers are devices used for programming and communicating with electronic devices that utilize FTDI chips for USB communication. An FTDI programmer typically includes a USB-to-serial converter chip and a connector to interface with the target device. These programmers enable easy and reliable data transfer and programming between a device and a computer. Commonly used for programming and debugging microcontrollers like Atmel AVR and Microchip PIC, FTDI programmers also serve applications requiring a USB-to-serial interface, such as GPS receivers and data loggers. They often feature voltage regulation, programmable clock sources, and other functionalities. FTDI programmers are essential tools for electronics designers and developers, providing a straightforward method for programming and communication. The typical pin configuration includes DTR (Data Terminal Ready), RXD (Receive Data), TXD (Transmit Data), VCC (Voltage Supply), CTS (Clear to Send), and GND (Ground). Each pin has its own specific function, such as resetting the target device, data transmission, and providing power and ground reference. It's important to consult the specific documentation for the exact pinout configuration.

CHAPTER 4. RESULTS AND DISCUSSION



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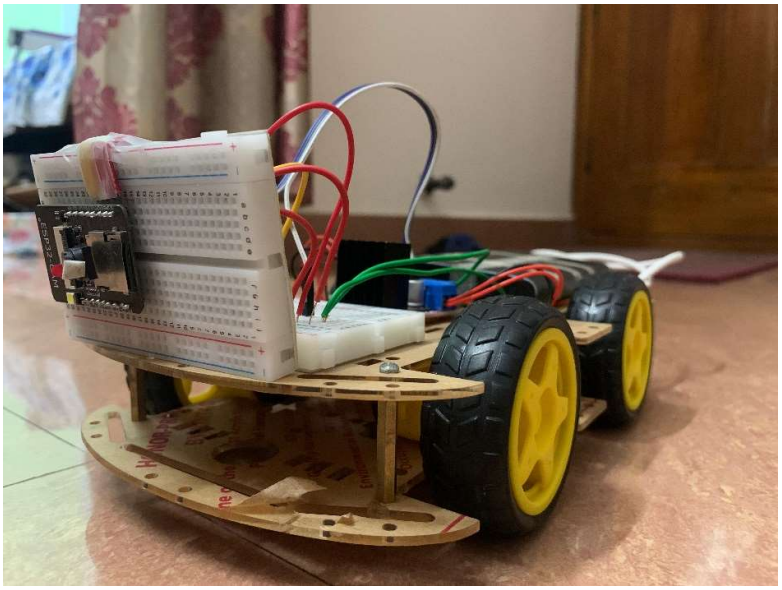


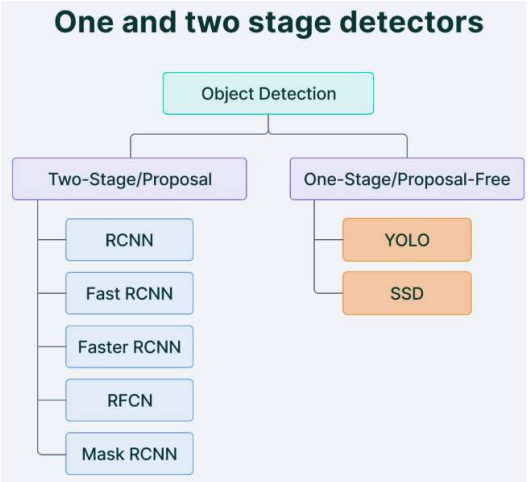
CHAPTER 5: WORKING ALOGORITHM FOR OBJECT DETECTION

Object detection is a popular task in computer vision.It deals with localizing a region of interest within an image and classifying this region like a typical image classifier. One image can include several regions of interest pointing to different objects. This makes object detection a more advanced problem of image classification. YOLO (You Only Look Once) is a popular object detection model known for its speed and accuracy. It was first introduced by Joseph Redmon et al. in 2016 and has since undergone several iterations, the latest being YOLO v9.
Object detection is a computer vision task that involves identifying and locating objects in images or videos. It is an important part of many applications, such as surveillance, self-driving cars, or robotics. Object detection algorithms can be divided into two main categories: single-shot detectors and two-stage detectors.One of the earliest successful attempts to address the object detection problem using deep learning was the R-CNN (Regions with CNN features) model, developed by Ross Girshick and his team at Microsoft Research in 2014. This model used a combination of region proposal algorithms and convolutional neural networks (CNNs) to detect and localize objects in images.Object detection algorithms are broadly classified into two categories based on how many times the same input image is passed through a network.

Two-shot object detection

Two-shot object detection uses two passes of the input image to make predictions about the presence and location of objects. The first pass is used to generate a set of proposals or potential object locations, and the second pass is used to refine these proposals and make final predictions. This approach is more accurate than single-shot object detection but is also more computationally expensive.

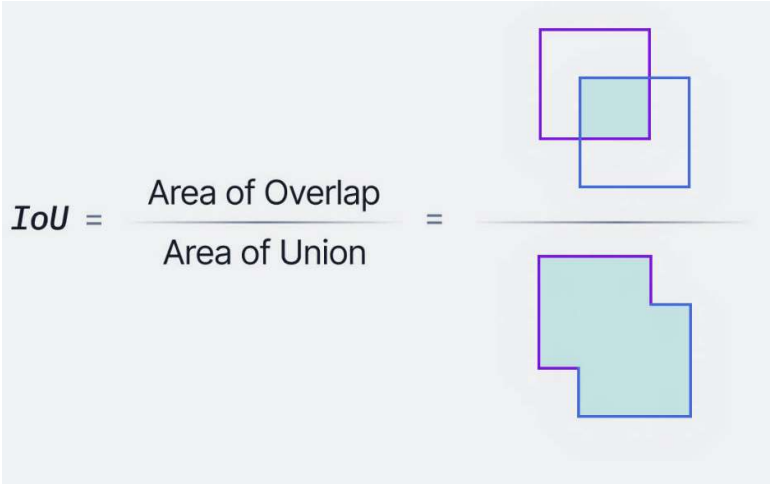
Overall, the choice between single-shot and two-shot object detection depends on the specific requirements and constraints of the application.Generally, single-shot object detection is better suited for real-time applications, while two-shot object detection is better for applications where accuracy is more important.



Object detection models performance evaluation metrics

To determine and compare the predictive performance of different object detection models, we need standard quantitative metrics.The two most common evaluation metrics are Intersection over Union (IoU) and the Average Precision (AP) metrics.Intersection over Union (IoU)Intersection over Union is a popular metric to measure localization accuracy and calculate localization errors in object detection models.

To calculate the IoU between the predicted and the ground truth bounding boxes, we first take the intersecting area between the two corresponding bounding boxes for the same object. Following this, we calculate the total area covered by the two bounding boxes—also known as the "Union" and the area of overlap between them called the "Intersection."The intersection divided by the Union gives us the ratio of the overlap to the total area, providing a good estimate of how close the prediction bounding box is to the original bounding box.



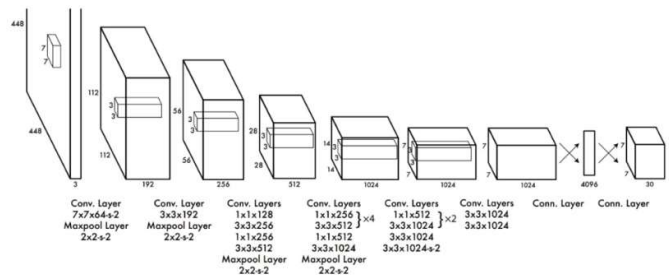
YOLO

You Only Look Once (YOLO) proposes using an end-to-end neural network that makes predictions of bounding boxes and class probabilities all at once. It differs from the approach taken by previous object detection algorithms, which repurposed classifiers to perform detection. Following a fundamentally different approach to object detection, YOLO achieved state-of-the-art results, beating other real-time object detection algorithms by a large margin.

While algorithms like Faster RCNN work by detecting possible regions of interest using the Region Proposal Network and then performing recognition on those regions separately, YOLO performs all of its predictions with the help of a single fully connected layer.Methods that use Region Proposal Networks perform multiple iterations for the same image, while YOLO gets away with a single iteration.

YOLO Architecture

The YOLO algorithm takes an image as input and then uses a simple deep convolutional neural network to detect objects in the image. The architecture of the CNN model that forms the backbone of YOLO is shown below.



The Architecture. Our detection network has 24 convolutional layers followed by 2 fully connected layers. Alternating 1×1 convolutional layers reduce the features space from preceding layers. We pretrain the convolutional layers on the ImageNet classification task at half the resolution (224×224 input image) and then double the resolution for detection.

The first 20 convolution layers of the model are pre-trained using ImageNet by plugging in a temporary average pooling and fully connected layer. Then, this pre-trained model is converted to perform detection since previous research showcased that adding convolution and connected layers to a pre-trained network improves performance. YOLO's final fully connected layer predicts both class probabilities and bounding box coordinates.

YOLO divides an input image into an S × S grid. If the center of an object falls into a grid cell, that grid cell is responsible for detecting that object. Each grid cell predicts B bounding boxes and confidence scores for those boxes. These confidence scores reflect how confident the model is that the box contains an object and how accurate it thinks the predicted box is.

YOLO predicts multiple bounding boxes per grid cell. At training time, we only want one bounding box predictor to be responsible for each object. YOLO assigns one predictor to be "responsible" for predicting an object based on which prediction has the highest current IOU with the ground truth. This leads to specialization between the bounding box predictors. Each predictor gets better at forecasting certain sizes, aspect ratios, or classes of objects, improving the overall recall score.

One key technique used in the YOLO models is non-maximum suppression (NMS). NMS is a post-processing step that is used to improve the accuracy and efficiency of object detection. In object detection, it is common for multiple bounding boxes to be generated for a single object in an image. These bounding boxes may overlap or be located at different positions, but they all represent the same object. NMS is used to identify and remove redundant or incorrect bounding boxes and to output a single bounding box for each object in the image.

CHAPTER 6. CHALLENGES FACED

- **DOWNLOADING DIFFERENT LIBRARIES :** One key challenge encountered was downloading and integrating various library functions required for the project. This involved identifying, obtaining, and incorporating libraries compatible with components like the ESP32-CAM module, and FTDI programmer
.
- Managing dependencies and resolving conflicts between libraries added complexity. Despite these hurdles, successfully integrating the necessary library functions was crucial for achieving the project's objectives.

- **Processing time:** This involved optimizing algorithms and code to ensure efficient execution within the required time frame. Despite the challenge, improving processing time was crucial for the project's success.

- **System requirements:** This involved identifying the necessary hardware and software specifications to ensure the smooth operation of the surveillance robot. Meeting these requirements was essential for the effective deployment and functionality of the system.

- **Voltage difference between the ESP-32 and the motor driver:** The ESP-32 cam required 5 V whereas the motor driver requires a minimum of 8 V to drive the motors. As we are using 4 motors and 2 servo motors it becomes challenging to regulate voltage between them.

- **Heat management system :** Within the few minutes of use (appprx. 3 minutes) the motor driver and ESP cam reaches an apprx. Of 75 degrees causing it to overheat and increases the delay time between the video stream. The heating of the motor driver causes motors to slow down hence reduces the efficiency of the motor.

CHAPTER 7. APPLICATIONS AND FUTURE ENHANCEMENT

7.1 General Applications of surveillance robot:

1. **Home Security**: The robot can patrol homes and alert owners to suspicious activity or intruders, enhancing security and peace of mind.

2. **Industrial Monitoring**: In industrial settings, the robot can monitor equipment, machinery, and facilities for faults, leaks, or other anomalies, helping to prevent accidents and downtime.

3. **Remote Area Surveillance**: It can be deployed in remote or hazardous locations to monitor wildlife, environmental conditions, or critical infrastructure, providing valuable data for research or security purposes.

4. **Emergency Response**: During emergencies such as natural disasters or search-and-rescue operations, the robot can navigate through hazardous environments to gather information or assist first responders.

5. **Public Safety**: In public spaces like parks or parking lots, the robot can help enhance safety by monitoring for suspicious behavior or emergencies and providing real-time alerts to authorities.

6. **Border Security**: Along borders or sensitive areas, the robot can patrol and monitor for illegal crossings or suspicious activity, aiding border security efforts.

7. **Traffic Management**: In urban areas, the robot can monitor traffic flow, detect accidents or congestion, and provide valuable data for optimizing transportation systems.

8. **Event Security**: During large events or gatherings, the robot can assist in crowd monitoring and security operations, helping to ensure the safety of attendees.

   Overall, the surveillance robot offers a cost-effective and efficient solution for monitoring and surveillance across various domains, contributing to enhanced safety, security, and efficiency.

7.2 Applications of the proposed model

- **Home Security:** The surveillance robot can monitor homes for security breaches, providing real-time video feeds and alerts to homeowners.
- **Industrial Monitoring:** It can be used in factories and warehouses to monitor operations, ensuring safety and security.
- **Remote Area Surveillance:** Ideal for monitoring remote or hard-to-reach locations, the robot can provide continuous surveillance without the need for human presence.
- **Pet Monitoring:** Homeowners can keep an eye on their pets while away, ensuring their safety and well-being.

7.3 Enhancement

Here are some potential areas for further development:

- **Advanced AI Integration:** Incorporating more sophisticated AI algorithms for better object detection and recognition.
- **Enhanced Mobility:** Improving the robot's mobility to navigate complex terrains and obstacles more effectively.
- **Longer Battery Life:** Developing more efficient power management systems to extend operational time.
- **Integration with IoT Devices:** Expanding compatibility with other IoT devices for a more comprehensive surveillance system.
- **Improved Connectivity:** Enhancing Wi-Fi and possibly adding cellular connectivity to ensure uninterrupted remote access

CHAPTER 8. CONCLUSION

In conclusion, the surveillance robot utilizing the ESP32-CAM module presents a promising solution for various surveillance and monitoring needs. The integration of the ESP32-CAM module, which offers a compact and cost-effective means of capturing images and streaming video, allows for a versatile application in both residential and industrial settings. By combining this module with a robotic platform, the system enables remote control and real-time video streaming, significantly enhancing its utility for home security, office monitoring, and remote area surveillance. The growing demand for efficient and affordable surveillance systems highlights the potential of this project to meet diverse monitoring requirements. This surveillance robot not only provides a practical approach to remote monitoring but also opens avenues for future enhancements, such as advanced AI integration, improved mobility, and extended battery life. Overall, the surveillance robot with ESP32-CAM is a valuable innovation, catering to the increasing need for reliable and accessible surveillance solutions.